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KPK BOARD TEXTBOOK NOTES - MATHEMATICS CLASS 9

Unit 8: Linear Graphs and Their Applications

Complete, easy English notes based on *A Textbook of Mathematics for Grade IX*, Khyber Pakhtunkhwa Textbook Board (KPTB), Peshawar - the Cartesian plane, plotting points, drawing straight-line graphs, reading graphs, conversion graphs, and real-life applications, with worked examples and exam practice.

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> **How to use these notes:** Always work with a pencil, a ruler and squared paper beside you. Copy every graph in these notes by hand once - in this unit, the graph itself is the answer, so neat plotting earns the marks. After each section, cover the worked answer and try the point or table yourself.

What this unit covers

1. The Cartesian plane - axes, origin and ordered pairs
2. The four quadrants and plotting and reading points
3. Drawing the graph of a linear equation from a table of values
4. Special lines: x = a, y = b, and lines through the origin
5. Reading values from a graph and solving two equations graphically
6. Conversion graphs, such as degrees Celsius and Fahrenheit
7. Applications: distance-time, total cost and pay graphs
8. Formula summary, common mistakes, MCQs and practice questions with answers

Source book: KPTB Mathematics for Grade IX, Unit 8 (book pages 192-209). The notes follow the book's order and syllabus. The scan used for verification carries no printed copyright page, so no printing year is claimed here - the 17-unit structure is the current circulating KPTB edition.

1. The Cartesian Plane and Ordered Pairs

Two number lines at right angles make the **Cartesian plane** (also called the coordinate plane). It lets us show the position of any point with just two numbers.

- The horizontal line is the **x-axis**. The vertical line is the **y-axis**.
- The point where the two axes cross is the **origin**, written **O = (0, 0)**.
- On the x-axis, numbers are positive to the right of the origin and negative to the left. On the y-axis, numbers are positive upwards and negative downwards.

Ordered pairs - x first, then y

The position of a point P is written as an **ordered pair (x, y)**. The first number, **x**, is the distance across from the origin. The second number, **y**, is the distance up (or down). The order matters: (2, 5) and (5, 2) are two different points.

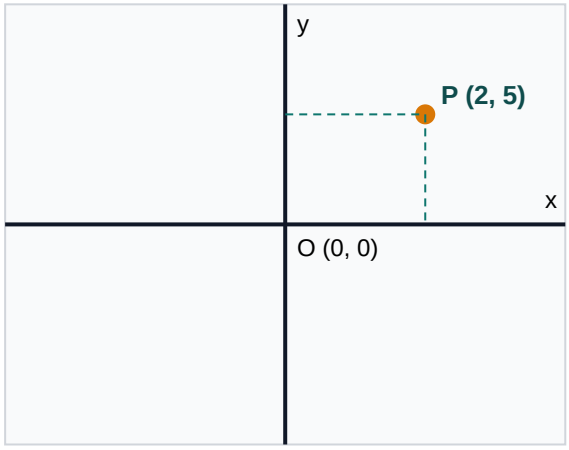


Figure 1. The point P (2, 5): 2 units across, then 5 units up.

Example 1. State the coordinates of a point 3 units to the left of the origin and 2 units down.
Answer: left means x is negative and down means y is negative, so the point is **(-3, -2)**.

Exam trap: Never swap the pair. Write x first, always. A point on the x-axis has y = 0, and a point on the y-axis has x = 0.

2. The Four Quadrants and Plotting Points

The two axes divide the plane into four parts called **quadrants**, numbered I, II, III and IV in the anti-clockwise direction, starting at the top right.

Quadrant	Position	Sign of x	Sign of y	Example point
I	Top right	+	+	(3, 4)
II	Top left	-	+	(-3, 4)
III	Bottom left	-	-	(-3, -4)
IV	Bottom right	+	-	(3, -4)

Points that lie exactly on an axis belong to **no quadrant**. For example, (5, 0) is on the x-axis and (0, -2) is on the y-axis.

How to plot a point - two moves only

1. Start at the origin.
2. Move **across** by the x value: right if positive, left if negative.
3. From there, move **up or down** by the y value: up if positive, down if negative. Mark the point and label it.

Example 2. In which quadrant does each point lie?

A (-2, 5): x negative, y positive - **Quadrant II**. B (4, -1): **Quadrant IV**. C (-6, -3): **Quadrant III**. D (0, 7): on the **y-axis**, in no quadrant.

Reading a point from a graph

Do the moves in reverse. From the point, drop straight down (or up) to the x-axis to read x, and go straight across to the y-axis to read y. Always write the pair as (x, y).

> **Memory line:** Across the landing, then up the stairs - x is the landing move, y is the stair move. Signs tell you the quadrant before you even plot.

3. Drawing the Graph of a Linear Equation

An equation such as y = 2x + 1 is called **linear** because x and y appear only to the power 1 - and its graph is always a **straight line**. Every point on that line is a solution of the equation, and every solution lies on the line.

The table method - four steps

1. Choose at least **three** easy values of x, for example -2, -1, 0, 1, 2. Zero is always worth including.
2. Put each x into the equation and calculate y. Write the results in a table.
3. Plot the three (or more) points.
4. Join them with a ruler and extend the line a little at both ends, with arrowheads. Label the line with its equation.

Example 3. Draw the graph of y = 2x + 1.

x	-2	-1	0	1	2
y = 2x + 1	-3	-1	1	3	5

Points: (-2, -3), (-1, -1), (0, 1), (1, 3), (2, 5). Plotted and joined, they form one straight line. The point where the line crosses the y-axis is (0, 1) - the constant term of the equation.

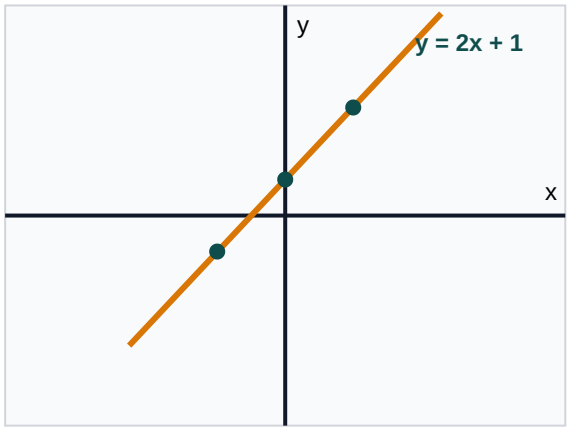


Figure 2. The straight-line graph of y = 2x + 1.

Why three points? Two points are enough to draw a line, but the third point is your check. If the three points do not lie in a straight line, one calculation is wrong - find it before you join them.

4. Special Straight Lines

Vertical and horizontal lines

Equation	What the line looks like	Key fact
x = a	A vertical line through (a, 0)	Every point on it has the same x. Example: x = 3 passes through (3, 0), (3, 2), (3, -1).
y = b	A horizontal line through (0, b)	Every point on it has the same y. Example: y = -2 passes through (0, -2), (4, -2), (-3, -2).
y = 0	The x-axis itself	Every point on the x-axis has y = 0.
x = 0	The y-axis itself	Every point on the y-axis has x = 0.

Lines through the origin: y = mx

If there is no constant term, the line passes through the origin. In **y = mx**, the number m controls the steepness: a bigger m makes a steeper line, and a negative m makes the line slope downwards from left to right.

Example 4. For y = 3x: when x = 0, y = 0; when x = 1, y = 3; when x = 2, y = 6. The line through (0, 0), (1, 3) and (2, 6) rises steeply. For y = -x + 4 the line slopes downwards, crossing the y-axis at (0, 4) and the x-axis at (4, 0).

The general line: y = mx + c

- **c** is where the line crosses the y-axis - the point (0, c). Read it straight from the equation.
- **m** is the steepness (gradient in later classes). Positive m: the line rises left to right. Negative m: it falls. m = 0 gives the horizontal line y = c.
- A quick sketch needs only two points: (0, c) on the y-axis, and one more point from a simple x value.

Check a point belongs to a line: Put its x and y into the equation. If both sides are equal, the point lies on the line. For example, (2, 5) lies on y = 2x + 1 because 2 x 2 + 1 = 5.

5. Reading a Graph and Solving Equations Graphically

Reading a value from a graph

To find y when x is given: start at the x value on the x-axis, go **straight up (or down)** to the line, then go **straight across** to the y-axis and read the value. To find x when y is given, do the same steps in the opposite order. Draw light dotted lines while you read - examiners award method marks for them.

Example 5. From the graph of y = 2x + 1 on page 4: when x = 1, the line gives y = 3. When y = 5, reading back across gives x = 2. Both readings match the table, as they must.

Solving two equations by their crossing point

Draw both lines on the same axes. The point where they **cross (intersect)** belongs to both lines, so its coordinates are the one pair (x, y) that satisfies both equations at the same time.

Example 6. Solve graphically: y = x + 1 and y = -x + 5.

For y = x + 1: points (0, 1), (2, 3), (4, 5). For y = -x + 5: points (0, 5), (2, 3), (4, 1).

The two lines cross at **(2, 3)**, so **x = 2, y = 3**. Check in both equations: 2 + 1 = 3 and -2 + 5 = 3. Both correct.

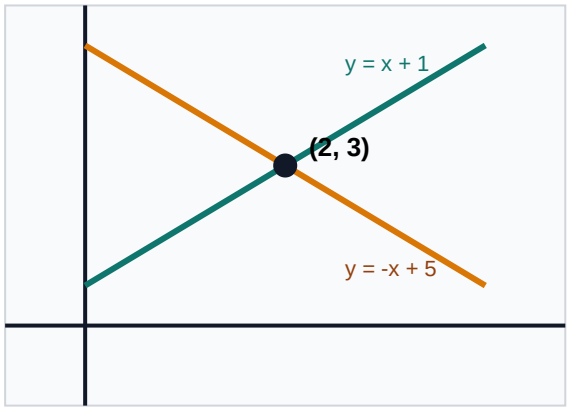


Figure 3. Two lines, one crossing point - that point is the common solution.

- If two lines are **parallel**, they never cross: the equations have **no common solution**.
- Always verify a graphical solution by substituting it back into both original equations.

6. Conversion Graphs

A **conversion graph** changes a quantity from one unit to another. Because the change is steady, the graph is a straight line - draw it once, then read off any conversion without calculating.

Celsius and Fahrenheit

F = (9/5) C + 32 Two easy points fix the line: water freezes at C = 0, F = 32, and boils at C = 100, F = 212.

Celsius (C)	0	20	40	60	80	100
Fahrenheit (F)	32	68	104	140	176	212

Example 7. Use the conversion line to change 30 degrees C into Fahrenheit, and 98.6 degrees F into Celsius.

From the graph (or the formula): at C = 30, F = (9/5 x 30) + 32 = 54 + 32 = **86 degrees F**. Reading the other way, F = 98.6 gives C = (98.6 - 32) x 5/9 = **37 degrees C** - normal body temperature.

Other everyday conversions

- **Currency:** rupees into dollars (or the reverse) - the line passes through the origin, because zero of one currency is zero of the other.
- **Length:** kilometres into miles - 5 miles is about 8 km, so the line through (0, 0) and (8 km, 5 miles) converts both ways.
- **Reading both ways:** the same graph converts forwards and backwards. Go up from one axis to the line, then across to the other axis.

Watch the intercept: The temperature line does **not** pass through the origin, because 0 degrees C is 32 degrees F, not 0. Currency and length lines do pass through the origin. Checking the starting point is the first step of every conversion question.

7. Applications: Distance, Cost and Pay

Distance-time graphs

When something moves at a steady speed, distance against time is a straight line through the origin: **distance = speed x time**. The steepness of the line is the speed - a steeper line means faster movement. A horizontal part means the object has stopped.

Example 8. A bus travels at a steady 40 km per hour. Distances: after 1 h, 40 km; after 2 h, 80 km; after 3 h, 120 km. The graph through (0, 0), (1, 40), (2, 80), (3, 120) lets you read, for example, that 100 km takes 2.5 hours.

Total-cost graphs: a fixed charge plus a rate

Many bills have a fixed starting charge plus a charge per unit. That gives the line **y = (rate) x + (fixed charge)**, which starts above the origin.

Example 9. A repair worker charges a fixed visit fee of Rs 300 plus Rs 50 per hour. Total cost y for x hours: y = 50x + 300.

Hours (x)	0	1	2	3	4
Cost in Rs (y)	300	350	400	450	500

The line starts at (0, 300) - the visit fee, paid even for zero hours - and rises by Rs 50 for each extra hour. From the graph, a bill of Rs 475 corresponds to 3.5 hours of work.

Pay and wages

Pay for hourly work, with no fixed amount, is a line through the origin: pay = (hourly rate) x (hours). A worker paid Rs 250 per hour earns Rs 1,000 in 4 hours and Rs 2,000 in 8 hours - the graph reads any hours-pay pair in seconds.

> **The pattern to recognise:** Starting value on the y-axis = the fixed amount (c). Steady rise = the rate per unit (m). Almost every application question in this unit is y = mx + c wearing real-life clothes.

8. Worked Mixed Examples

Example 10. Draw the graph of y = -x + 4 and write down where it crosses each axis.

Table: x = 0 gives y = 4; x = 2 gives y = 2; x = 4 gives y = 0. The line through (0, 4), (2, 2), (4, 0) slopes downwards. It crosses the **y-axis at (0, 4)** and the **x-axis at (4, 0)**. Quick rule: to find the x-axis crossing, put y = 0 in the equation: 0 = -x + 4, so x = 4.

Example 11. A tank holds 20 litres of water. A tap adds 5 litres per minute. Draw the graph of the water in the tank and find when it reaches 70 litres.

Amount y after x minutes: y = 5x + 20. Points: (0, 20), (4, 40), (8, 60), (10, 70). Reading the graph at y = 70 gives x = 10, so the tank reaches 70 litres after **10 minutes**. The intercept 20 is the water already in the tank; the rate 5 is the steepness.

Example 12. Does the point (3, 10) lie on the line y = 3x + 2?

Put x = 3 into the equation: y = 3 x 3 + 2 = 11, not 10. So **(3, 10) does not lie on the line** - the point on the line with x = 3 is (3, 11).

Choosing scales - a practical skill

- Look at the biggest x and y values first, then choose a scale so the graph fills most of the page: for example, 1 cm = 10 units for large values.
- The two axes may use **different** scales - that is allowed - but each axis must keep one steady scale along its whole length.
- Label both axes with the quantity and its unit in application graphs, for example "Time (hours)" and "Distance (km)".

9. Formula Summary and Common Mistakes

Idea	Rule
Ordered pair	(x, y) - x across first, then y up or down. Origin is (0, 0).
Quadrants	I (+, +), II (-, +), III (-, -), IV (+, -). Points on an axis are in no quadrant.
Linear equation	Graphs as a straight line. Plot at least 3 points from a table as a check.
y = mx + c	c = crossing point on the y-axis, (0, c). m = steepness; negative m slopes downwards.
x = a and y = b	x = a is vertical; y = b is horizontal.
X-axis crossing	Put y = 0 in the equation and solve for x.
Graphical solution	Two equations are solved at the crossing point of their two lines.
Conversion graph	F = (9/5) C + 32. Fixed points: 0 C = 32 F and 100 C = 212 F.
Applications	Distance = speed x time. Total cost = rate x quantity + fixed charge.

Five mistakes that lose marks

1. Writing the pair backwards - (y, x) instead of (x, y).
2. Joining points freehand or with a curve instead of a ruler-straight line.
3. Plotting only two points, so one wrong calculation is never caught.
4. Starting a conversion graph at the origin when it should start at an intercept, like 32 degrees F.
5. Forgetting to label the axes, the units, and the line with its equation.

> **One-line revision:** Across then up to plot, table then line to draw, up then across to read - and the crossing point solves two equations at once.

10. Practice - MCQs and Questions

Multiple-choice questions

1. The point (0, 5) lies on: (a) the x-axis (b) the y-axis (c) Quadrant I (d) Quadrant IV
2. The point (-3, 4) lies in: (a) Quadrant I (b) Quadrant II (c) Quadrant III (d) Quadrant IV
3. The graph of x = 3 is a line: (a) parallel to the x-axis (b) parallel to the y-axis (c) through the origin (d) sloping downwards
4. On the graph of y = 3x, when x = 2, y equals: (a) 3 (b) 5 (c) 6 (d) 9
5. The coordinates of the origin are: (a) (1, 0) (b) (0, 1) (c) (1, 1) (d) (0, 0)
6. The line y = x + 2 crosses the y-axis at: (a) (2, 0) (b) (0, 2) (c) (-2, 0) (d) (0, -2)

Short and long questions

1. Plot the points A (2, 3), B (-1, 4), C (-3, -2), D (4, -3) and state the quadrant of each.
2. Draw the graph of y = 2x - 3 for x from -1 to 3, and read from your graph the value of y when x = 2.5.
3. Solve graphically: y = x + 2 and y = -x + 6.
4. Using a conversion graph or the formula, change 50 degrees C to Fahrenheit and 77 degrees F to Celsius.
5. A car travels at a steady 60 km per hour. Draw a distance-time graph for the first 4 hours and use it to find the time taken for 210 km.

Answer key

MCQs: 1(b), 2(b), 3(b), 4(c), 5(d), 6(b).

Questions: 1) A Quadrant I, B Quadrant II, C Quadrant III, D Quadrant IV. 2) Points (-1, -5), (0, -3), (1, -1), (2, 1), (3, 3); at x = 2.5, y = 2. 3) Lines cross at (2, 4), so x = 2, y = 4. 4) 50 C = 122 F; 77 F = 25 C. 5) Distances 0, 60, 120, 180, 240 km; 210 km takes 3.5 hours.

> **Final tip:** In the board paper, show the table, plot with a sharp pencil, join with a ruler, and label everything. In graph questions, neat and labelled work is where most of the marks are.